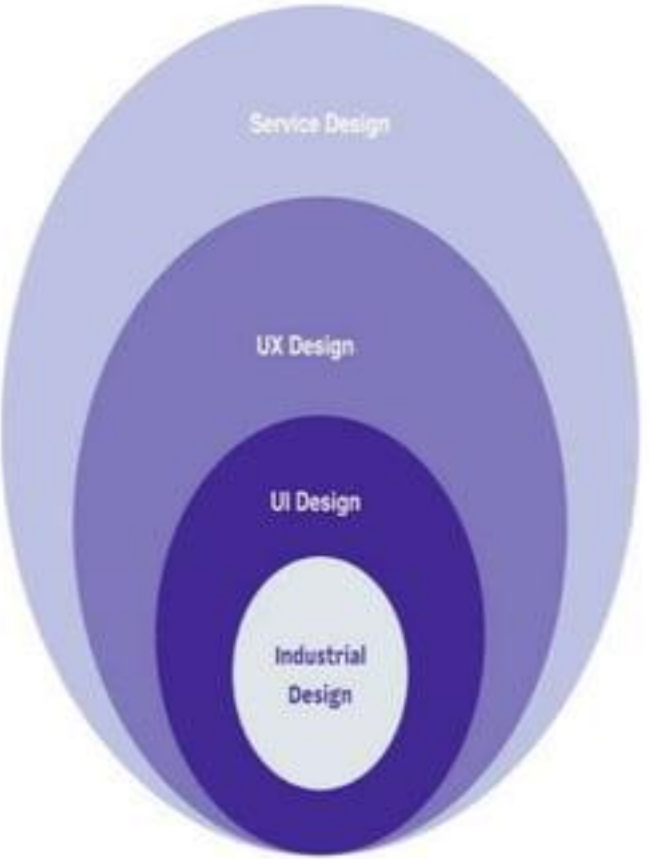
7 Design principles for IoT

For designers focused on designing SW services and screen based interfaces or physical products, designing IoT solutions creates totally new design challenges. IoT solutions consist of multiple elements: physical devices like sensors, actuators and interactive devices, the network connecting these devices, the data gathered from these devices and analyzed to create a meaningful experience and last but definitely not least, the physical context in which user interacts with the solution. You need to do various types of design, from industrial product design to service and business design. All of these factors have their impact to the total UX of the IoT system and the task of designing in this context may feel quite overwhelming. To make it a little easier, I have gathered my list of the 7 most important design principles for IoT.

Layers of design for meaningful IoT experience_



Design principles for IOT

1. Focus on value
2. Take a holistic view
3. Put safety first
4. Consider the context
5. Build a strong brand
6. Prototype early and often
7. Use data responsibly

1. Focus on value

In the world of IoT, user research and service design are more crucial than ever. While early adopters are eager to try out new technology, many others are reluctant to take new technology into use and cautious about using it, due to not feeling confident with it. For your IoT solution to become widely adopted, you need to dig deep into users' needs in order to find out where lies a problem truly worth solving and what is the real end user value of the solution. You also need to understand what might be the barriers of adopting the new technology in general and your solution specifically. For deciding on your feature set, you need research too. The features that might be valuable and highly relevant for the tech early adopters may be uninteresting for the majority of the users and vice versa, so you need to plan carefully what features to include and in which order.

2. Take a holistic view

IoT solutions typically consist of multiple devices with different capabilities and both physical and digital touchpoints. The solution may also be provided in co-operation with multiple different service providers. It is not enough to design one of the touchpoints well, instead you need to take a holistic look across the whole system, the role of each device and service, and the conceptual model of how user understands and perceives the system. The whole system needs to work seamlessly together in order to create a meaningful experience.

3. Put safety first

As the IoT solutions are placed in the real world context, the consequences can be serious, when something goes wrong. At the same time the users of the IoT solutions may be vary of using new technology, so building trust should be one of your main design drivers. Trust is built slowly and lost easily, so you really need to make sure that every interaction with the product/service builds the trust rather than breaks it. What it means in practise? First of all, it means understanding possible error situations related to context of use, HW, SW and network as well as to user interactions and trying to prevent them. Secondly, if the error situations still occur, it means appropriately informing the user about them and helping them to recover. Secondly, it means considering data security & privacy as key elements of your design. It is really important for users to feel, that their private data is safe, their home, working environment and everyday objects cannot be hacked and their loved ones are not put at risk. Thirdly, quality assurance is critical and it should not only focus on testing the SW, but on testing the end to end system, in a real-world context.

4. Consider the context

IoT solutions exist at the crossroads of the physical and digital worlds. Commands given through digital interfaces may produce real world effects, but unlike digital commands, the actions happening in the real-world cannot necessarily be undone. In the real world context lots of unexpected things can happen and at the same time user should be able to feel safe and in control. The context places also other kind of requirements to the design. Depending on the physical context, the goal might be to minimize distraction of the user or e.g. to design devices that hold up against changing weather conditions. IoT solutions in homes, workplaces and public areas are are typically multi-user systems and thus less personal than e.g. screen based solutions used in smartphones, which also brings into picture the social context where the solution is used and its' requirements for the design.

5. Build a strong brand

Due to the real world context of the IoT solutions, regardless of how carefully you design things and aim to build trust, something unexpected will happen at some point and your solution is somehow going to fail. In this kind of situations, it is of utmost importance, that you have built a strong brand that truly resonates with the end users. When they feel connected to your brand, they will be more forgiving about the system failures and will still keep on using your solution. While designing your brand, you must keep in mind, that trust should be a key element of the brand, one of the core brand values. This core value should also be reflected in the rest of the brand elements, like the choice of color, tone of voice, imagery etc.

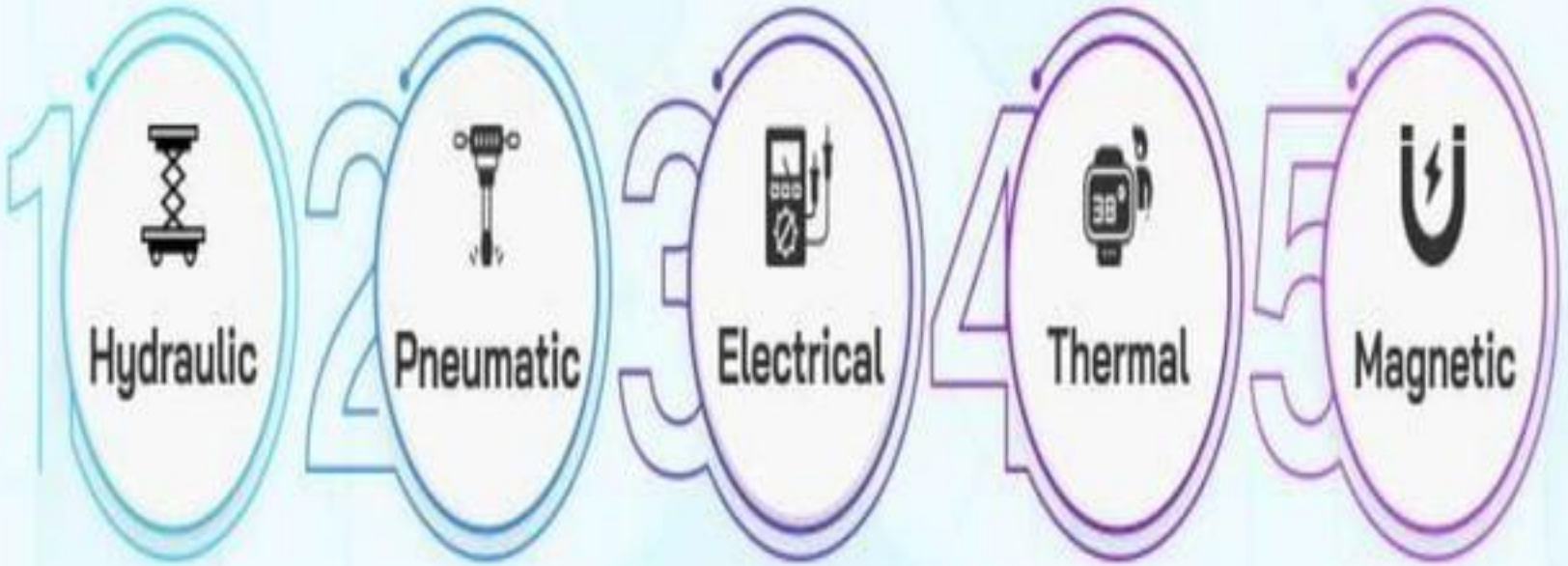
6. Prototype early and often

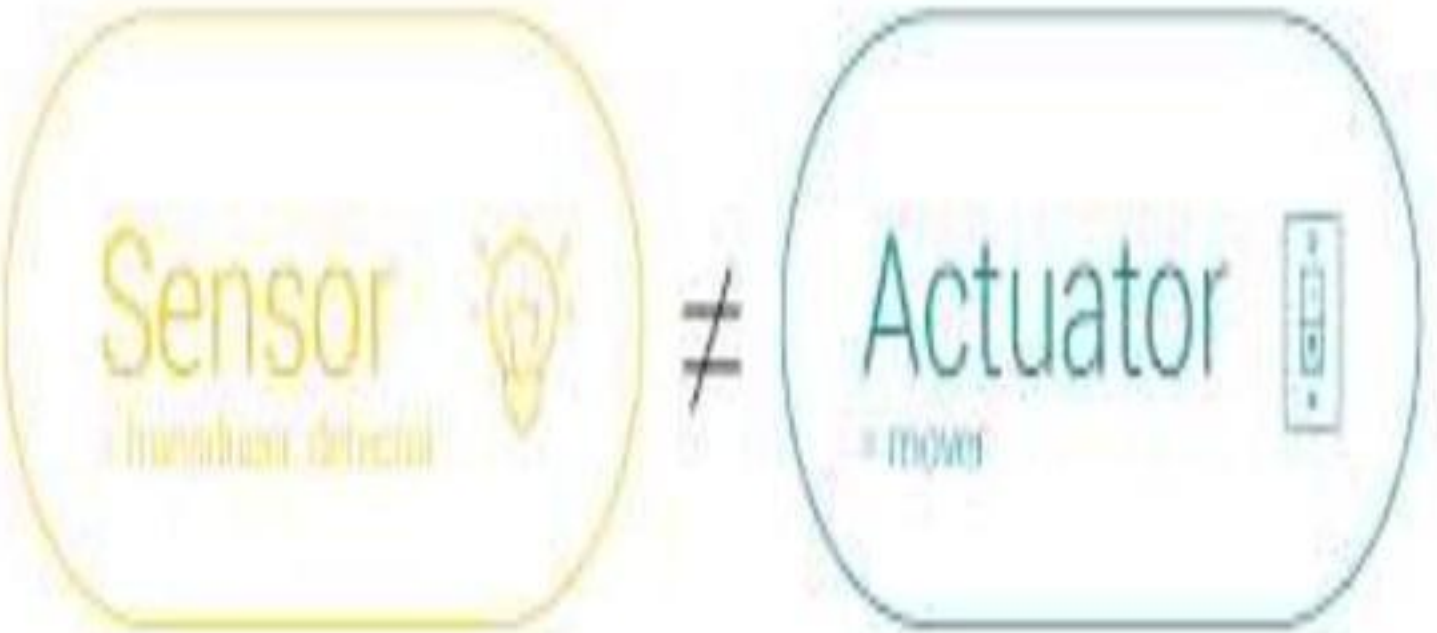
Typically HW and SW have quite different lifespans, but as successful IoT solution needs both the HW and SW elements, the lifespans should be aligned. At the same time, IoT solutions are hard to upgrade, because once the connected object is placed somewhere, it is not so easy to replace it with a newer version, especially if the user would need to pay for the upgrade and even the software within the connected object may be hard to update due to security and privacy reasons. Due to these factors and to avoid costly hardware iterations, it's crucial to get the solution right, from the beginning of implementation. What this means from the design perspective is that prototyping and rapid iteration of both the HW and the whole solution are essential in the early stages of the project. New, more creative ways of prototyping and faking the solution are needed.

7. Use data responsibly

IoT solutions can easily generate tons of data. However, the idea is not to hoard as much data as possible, but instead to identify the data points that are needed to make the solution functional and useful. Still, the amount of data may be vast, so it's necessary for the designer to understand the possibilities of data science and how to make sense of the data. Data science provides a lot of opportunities to reduce user friction, i.e. reducing use of time, energy and attention or diminishing stress. It can be used to automate repeated context dependent decisions, to interpret intent from incomplete/inadequate input or to filter meaningful signals from noise. Understanding what data is available and how it can be used to help the user is a key element in designing successful IoT services.

Types of Actuators




Sensor
≠
Actuator

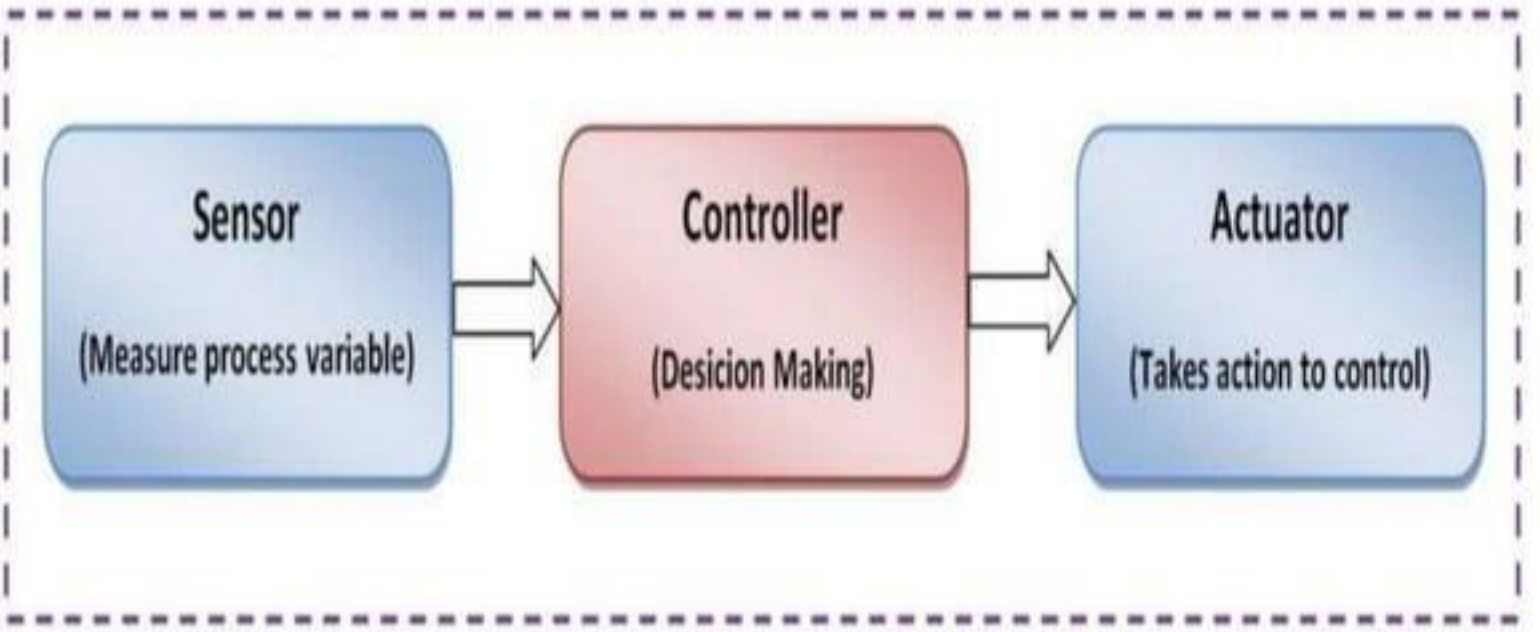
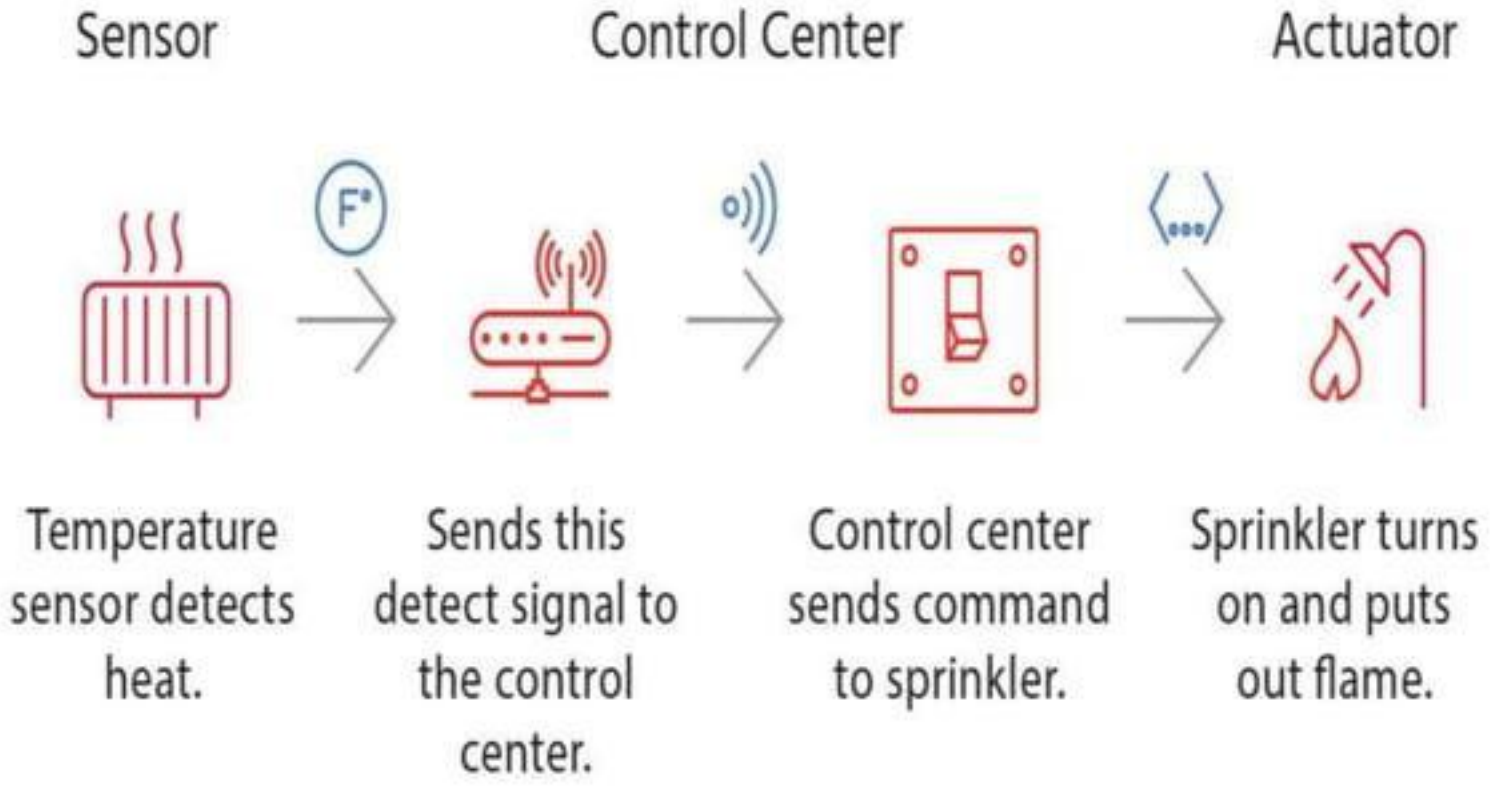


Figure- Sensor and actuator in a system



Sensor to **Actuator** Flow

Needed Capabilities - IOT

Needed Capabilities - IOT

1. Connectivity
2. Control
3. Device Management
4. Actionable Data

1. Connectivity

It starts with how a device or sensor connects to the internet and a cloud platform. There are many options to choose from WiFi through a hub or gateway, 2G, or 3G cellular networks. Once you have connectivity in place, now you can get the device or sensor talking to your cloud IoT platform. Ensure you find a service provider that can send data through clean API's that are easy to implement and install. This will ensure you can get quickly setup and start capturing your data within minutes.

2. Control

The next capability necessary when evaluating an IoT data platform is control of the device. There are a number of different scenarios for control including controlling a device through an application, device-to-device communication, or control from the cloud (based on an event, rule or some other pre-determined condition). For example, if you have a water leak detector, it can automatically send a command to the device which could be an appliance or part of the core infrastructure to turn off the water valve. Here, using two-way communication, a signal can be sent from the detector to the device via the cloud to shut off the water. Lastly, you can program the device from an app (or website) to shut off at a certain time or schedule based on a pre-programmed rule.

3. Device Management

Device management is also a major consideration. To keep devices and sensors up to date and functional, a strong device management solution is a core component of an IoT cloud platform. There are a few main capabilities a device management platform provides, including the ability for manufacturers to send software or firmware updates OTA (over-the-air), factory provisioning, as well as an out-of-box experience (OOBE).OOBE is part of a core experience that is often left to the last minute or completely glossed over. It's the first experience that an end user, be it a consumer, installer or technician has when interacting with a device for the first time. A great OOBE experience significantly increases the probability of an end user successfully installing and configuring a device. Furthermore, it reduces the likelihood of a support call or the end user returning the product altogether.

4. Actionable Data

The last capability you should consider in a IoT data platform is how you can query the data in a manner that is clear and meaningful. It's one thing to get all your data in place, but the value of the data is only realized when it's turned into information that can help solve a problem. We want organizations to focus on their core competency, like making great appliances or services that deliver value to their customers, rather than focusing on cloud infrastructure that makes it possible. At Buddy, our job is to provide an end-to-end turn-key solution to connect the world's IoT devices and provide real-time business insights for decision making. That can take the form of simple dashboard or deep analytics through integration with partners and services.

CAPABILITIES OF IOT

IoT capability	Description
Monitoring	Sensors and external data sources enable the comprehensive monitoring of: • the product's condition • the external environment • the product's operation and usage Monitoring also enables alerts and notifications of changes
Control	Software embedded in the product or in the product cloud enables: • Control of product functions • Personalization of the user experience
Optimization	Monitoring and control capabilities enable algorithms that optimize product operation and use in order to: • Enhance product performance • Allow predictive diagnostics, service, and repair
Autonomy	Combining monitoring, control, and optimization allows: • Autonomous product operation • Self-coordination of operation with other products and systems • Autonomous product enhancement and personalization • Self-diagnosis and service